

28 *Use of the Data Booklet is relevant to this question.*

An electrochemical cell is set up using a $\text{Zn}^{2+}(\text{aq})|\text{Zn}(\text{s})$ half-cell and a $\text{MnO}_4^{-}(\text{aq})|\text{Mn}^{2+}(\text{aq})$ half-cell in acidic solution.

Which of the following gives a correct effect on the E_{cell} when each of the changes is made to the corresponding half-cell separately?

	change	half-cell	effect on E_{cell}
A	addition of water	$\text{MnO}_4^{-}(\text{aq}) \text{Mn}^{2+}(\text{aq})$	less positive
B	addition of $\text{Mn}(\text{NO}_3)_2(\text{s})$	$\text{MnO}_4^{-}(\text{aq}) \text{Mn}^{2+}(\text{aq})$	no change
C	addition of $\text{Zn}(\text{NO}_3)_2(\text{s})$	$\text{Zn}^{2+}(\text{aq}) \text{Zn}(\text{s})$	more positive
D	addition of $\text{Na}_2\text{CO}_3(\text{s})$	$\text{Zn}^{2+}(\text{aq}) \text{Zn}(\text{s})$	no change